

Department of Mechanical Engineering

Course Outcomes of all courses of B Tech 4th semester MECHANICAL ENGG.

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C211 Applied Thermodynamics B037411(037)	C 211.1	Analyze and evaluate gas power cycles. (Level 4)
	C 211.2	Analyze reciprocating air compressors. (Level 4)
	C 211.3	Analyze vapour power cycle. (Level 4)
	C 211.4	Analyze steam condenser and discuss working principle of cooling pond and cooling towers. (Level 4,1)
	C 211.5	Analyze thermodynamic system with compressible fluid. (Level 4)

On successful completion of this course, students should be able to:

Course	COURSE OUTCOMES	
C212- Fluid Mechanics B037412(037)	C212.1	Explain fluid properties and basic principles of fluid statics and analyze the problem related to manometry, forces on submerge plane, buoyancy and flotation. (Level 1,3,)
	C212.2	Explain basic principles of fluid kinematics and analyze related practical problem. (Level 1,3,)
	C212.3	Explain basic principles of fluid dynamics and analyze related practical problem. (Level 1,3,)
	C212.4	Derive relationships for various flow characteristics of laminar flow, turbulent flow and energy losses in pipe flow and apply to analyze related practical problems. (Level 3)
	C212.5	Apply dimensional analysis to derive a relationship among connected variables and apply model laws to predict the behavior of the prototype in given circumstances. (Level 3)

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Course	COURSE OUTCOMES	
C213- Strength of Materials B037413(037)	C213.1	Apply the concept of stress and strain to analyze various types of structures. (Level 1,3,)
	C213.2	Determine the distribution of shear force, bending moment and transverse shear stress along the loaded beam. (Level 4,)
	C213.3	Determine the deflections and slope of loaded flexural members. . (Level 4,)
	C213.4	Analyze shaft and springs under torsional load. (Level 3)
	C213.5	Analyze various structural elements subjected to combined stresses/combined loads. (Level 3,)

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Course	COURSE OUTCOMES	
C214-Manufacturing Process B037414(037)	C214.1	Describe various metal casting and allied processes. (Level 2)
	C214.2	Describe various arc and gas welding processes. (Level 2)
	C214.3	Describe resistance welding, other special type of welding , soldering, brazing and braze welding (Level 2)
	C214.4	Describe construction, working and various machining operations of lathe, shaper and planer (Level 2)
	C 214.5	Describe construction, working and various machining operations of milling, broaching, drilling rimming and boring machine (Level 2)

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Course	COURSE OUTCOMES	
C215- Kinematic of Machine B037415(037)	C215.1	Describe the concepts of machines, mechanisms and related terminologies and analyze planar mechanism for displacement and velocity. (Level 2,3)
	C215.2	Analyze planar mechanism for acceleration. (Level 3)
	C215.3	Analyze cam-follower mechanism. (Level 3)
	C215.4	Analyze gears and gear train. (Level 3)
	C215.5	Analyze bearings, belt-drive, brakes and dynamometer. (Level 3)

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Course	COURSE OUTCOMES	
C216- Fluid Mechanics Lab B037421(037)	C216.1	Demonstrate practical understanding of principles of buoyancy and flotation and determine meta-centric height. (Level 3,5)
	C216.2	Verify impulse momentum principle (Level 5)
	C216.3	Demonstrate practical understanding of the various terms in Bernoulli's equation and verify Bernoulli's theorem. (Level 3,5)
	C216.4	Calibrate flow measurement devices (Level 3)
	C216.5	Demonstrate practical understanding of Major and Minor Losses in pipe flow. (Level 3)

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Course	COURSE OUTCOMES	
C217- Material Testing Lab B037422(037)	C217.1	Analyze mechanical properties of various engineering materials under specific types of load in universal testing machine. (Level 3)
	C217.2	Analyze mechanical properties of engineering materials under impact loading. (Level 3)
	C217.3	Analyze mechanical properties of specimen under torsion (Torsion Testing Machine, Spring Testing Machine) (Level 3)
	C217.4	Determine hardness of given material. (Level 3)
	C217.5	Analyze mechanical properties of specimen under fatigue, deep drawing and buckling load. (Level 3)

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Course	COURSE OUTCOMES	
C218- Manufacturing Process Lab B037423(037)	C218.1	Demonstrate the use of green sand molding process for casting. (Level 3)
	C218.2	Demonstrate the use of various machine tools for important machining operations. (Level 3)
	C218.3	Explain the tool geometry of single point cutting tool and twist drill. (Level 2)
	C218.4	Explain the practicability of various metal joining processes like arc welding, resistance welding, soldering and brazing. (Level 2)
	C218.5	Obtain practical skills in inspection and testing of casting and welding defects. (Level 3,4)



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Course	COURSE OUTCOMES	
C219- Virtual Lab B037424(037)	C219.1	Perform experiments of material testing laboratory through virtual simulator. (Level 3)
	C219.2	Analyze different type of mechanism through virtual simulator. (Level 3)
	C219.3	Analyze various heat transfer parameter in virtual laboratory (Level 3)
	C219.4	Describe EDM, Laser cutting, ECM after learning the process through micromachining laboratory. (Level 2)
	C219.5	Describe casting/ 3D scanning after learning the process through fabrication laboratory. (Level 2)